

PaddyGuard AI: A Multimodal AI Based System for Rice Leaf Disease & Pest Detection, Treatment Recommendation and Knowledge Assistance

1. Voice-Based Rice Disease Diagnosis using Speech Recognition and Natural Language Processing

Small Description

This component allows farmers to describe observed symptoms using voice. The system converts speech into text, extracts symptom keywords, and predicts the rice disease category (Bacterial Blight, Leaf Blast, Brown Spot, Healthy). The response is returned as both text and synthesized speech to support low-literacy users.

Research Questions

- Can natural language symptom descriptions accurately identify rice diseases?
- Does transformer-based NLP outperform traditional ML text classification?
- How reliable is voice-based diagnosis compared to image-based detection?

Technologies

- Whisper/Vosk etc...
- TF-IDF + SVM/Naive Bayes
- BERT
- Text-to-Speech
- PyTorch/Transformers

Implementation Steps (Brief)

- Collect symptom description data and create a dataset
- Convert voice to text
- Clean and normalize text
- Extract features
- Train NLP classification model
- Evaluation
- Convert prediction → voice output

2. Rice Leaf Disease Classification with Out-of-Distribution Detection using Deep Learning

Small Description

This module detects rice leaf diseases from images using deep learning and identifies unknown samples using OOD detection to prevent incorrect predictions when non-rice or unseen diseases are uploaded. The system is going to train for predicting the rice disease category (Bacterial Blight, Leaf Blast, Brown Spot, Healthy).

Research Questions

- Which CNN architecture performs best for rice disease classification?
- Can OOD detection reduce false predictions on unseen images?
- How does confidence calibration affect model reliability?
- Does transfer learning improve generalization in field conditions?

Technologies

- ResNet50 / EfficientNet / DenseNet etc...
- Transfer Learning
- Data Augmentation
- ODIN / OpenMax / Mahalanobis OOD detection
- PyTorch / TensorFlow

Implementation Steps (Brief)

- Preparing dataset
- Split train/val/test
- Train CNN models
- Evaluate performance
- Implement OOD detection
- Tune thresholds
- Generate explainability maps
- Export trained model

3. Hybrid AI and Knowledge-Based Rice Treatment Recommendation & Advisory Chatbot

Small Description

This component provides treatment recommendations based on disease type and integrates a chemical dosage calculator and a RAG-based agricultural chatbot that combines traditional and modern treatment knowledge.

Research Questions

- Can hybrid AI + rule-based recommendation improve treatment accuracy?
- How effective is a domain-specific RAG chatbot in agricultural advisory?
- Can dosage calculators reduce pesticide misuse?
- Does knowledge retrieval improve farmer decision confidence?

Technologies

- Rule-based expert system
- Retrieval Augmented Generation (RAG)
- Vector database (FAISS / Chroma DB)
- LLM (Llama / OpenAI compatible)
- MongoDB / PostgreSQL
- Prompt engineering

Implementation Steps (Brief)

- Create dosage calculator
- Build treatment knowledge base using Collected documents
- Define rule engine
- Create embeddings
- Build vector database
- Implement RAG pipeline
- Integrate chatbot UI
- Connect with disease prediction outputs

4. Rice Pest Detection using Deep Learning with Out-of-Distribution Detection

Small Description

This module detects common rice pests from images using Deep Learning architectures and uses OOD detection methods to identify unknown insects that are not part of the trained dataset, improving field reliability.

Research Questions

- Can CNN models accurately identify rice pests under real conditions?
- How effective is OOD in pest detection?
- Does background complexity affect detection performance?
- Can pest detection complement disease diagnosis in decision support?

Technologies

- ResNet50 / EfficientNet / DenseNet/ YOLOv8 etc..
- OOD Detection
- Image preprocessing & augmentation
- PyTorch / Tensorflow

Implementation Steps (Brief)

- Prepare pest dataset
- Train detection/classification model
- Evaluate accuracy
- Implement OOD detection
- Tune thresholds
- Export trained model